

J-Anne Yow

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Summary

Final-year PhD candidate in Robotics & AI at NTU Singapore, developing adaptive robotic systems that integrate reinforcement learning, multimodal perception, and language-driven control. Experienced in sim-to-real transfer, ROS/ROS2, and real-world deployment in user trials. Passionate about bridging research and application by enabling embodied AI models to transfer from simulation into practical, deployable robots.

Education

Nanyang Technological University, Singapore

Sept 2021 – Feb 2026

Ph.D. in Mechanical Engineering

(Expected)

- **Supervisor:** Prof Ang Wei Tech
- **Dissertation:** Towards Personalized Robot Assistance: Integrating User Preferences in Robot-Assisted Feeding
- **Research Interests:** Human-Robot Interaction, Machine Learning, Foundation Models, Continual Learning

Nanyang Technological University, Singapore

Aug 2017 – May 2021

BE in Mechanical Engineering with a Second Major in Business

- **GPA:** 4.92/5.00, Dean's List every Academic Year
- **Awards:** Dr Leung Shiu Kee Gold Medal Award, ASEAN Undergraduate Scholarship

Research Experience

Embodied AI Data Factory - Improving Teleoperation for Scalable Data Collection

2025 – present

- Leading efforts to enhance teleoperation for bimanual and dexterous manipulation to reduce cognitive load and preserve natural human motion.
- Building a scalable pipeline with markerless motion capture, multisensory perception, and real-time retargeting to dexterous hands, manipulators, and humanoids in simulation and on real hardware.
- Goal: To close the gap between human and robot motion to enable efficient, high-fidelity datasets for training robust, generalizable, foundation model-based vision-language-action (VLA) policies.

Robot-Assisted Feeding

2022 – present

- Led the technical delivery of an end-to-end adaptive feeding system, integrating perception, planning, and control into a safe, deployable platform for real-world elderly user trials.
- Developed goal-conditioned reinforcement learning policies in MuJoCo, integrating multimodal feedback (vision + force-torque sensing) for adaptive manipulation across variable food properties, achieving successful sim-to-real transfer.
- Built a language-driven adaptation pipeline, leveraging large language models to map user feedback into motion parameters for reliable, safe robot behaviour.

Grasping in Clutter

2020 – 2023

- Developed a point-and-click interface for human-in-the-loop robotic manipulation, integrating object segmentation, grasp ranking, and grasp cycling to streamline user input and improve selection efficiency in cluttered scenes.
- Developed a POMDP-based shared autonomy framework that queries users under intent uncertainty, reducing conflicting actions and enabling faster task completion.

Work Experience

Research Assistant, Nanyang Technological University – Singapore

Aug 2021 – present

- Managed ROS/ROS2-based system integration and hardware-software interfacing, collaborating with research staff and students to define technical goals and evaluate prototypes in simulation and on physical robots.
- Managed project planning, resource coordination, and ethical compliance, removing operational bottlenecks such as compute limitations to keep development on schedule.

- Mentored and scoped eight undergraduate projects, defining research objectives, guiding technical execution and skill development.
- Drafted and contributed to competitive funding proposals to support ongoing research and new initiatives.

Selected Publications

SAVR: Scooping Adaptation for Variable food properties via Reinforcement Learning	Oct 2025
<i>J-Anne Yow</i> , Wei Tech Ang	
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	
FRANC: Feeding Robot for Adaptive Needs and Personalized Care	Oct 2025
<i>J-Anne Yow</i> , Luke Toh, Yi Heng San, Wei Tech Ang	
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	
ORBiT: Optimizing Robot-Assisted Bite Transfer Leveraging a Real2Sim2Real Framework	Oct 2025
Sherwin Chan, <i>J-Anne Yow</i> , Yi Heng San, Vasanth Ravichandram, Yifan Wang, Lek Syn Lim, Wei Tech Ang	
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	
Shared Autonomy of a Robotic Manipulator for Grasping under Human Intent Uncertainty using POMDPs	Nov 2023
<i>J-Anne Yow</i> , Neha P Garg, Wei Tech Ang	
IEEE Transactions on Robotics (T-RO)	

Service

Reviewer: IEEE Robotics and Automation Letters (RAL), IEEE International Conference on Robotics and Automation (ICRA)

Skills

Programming: Python, C/C++

Robotics & Simulation: ROS/ROS2, MoveIt, MuJoCo, Isaac Lab

Machine Learning: Reinforcement Learning, Imitation Learning, PyTorch

Perception & Control: Vision, Force-Torque Sensing, Impedance/Admittance Control

Languages: English, Chinese, Malay